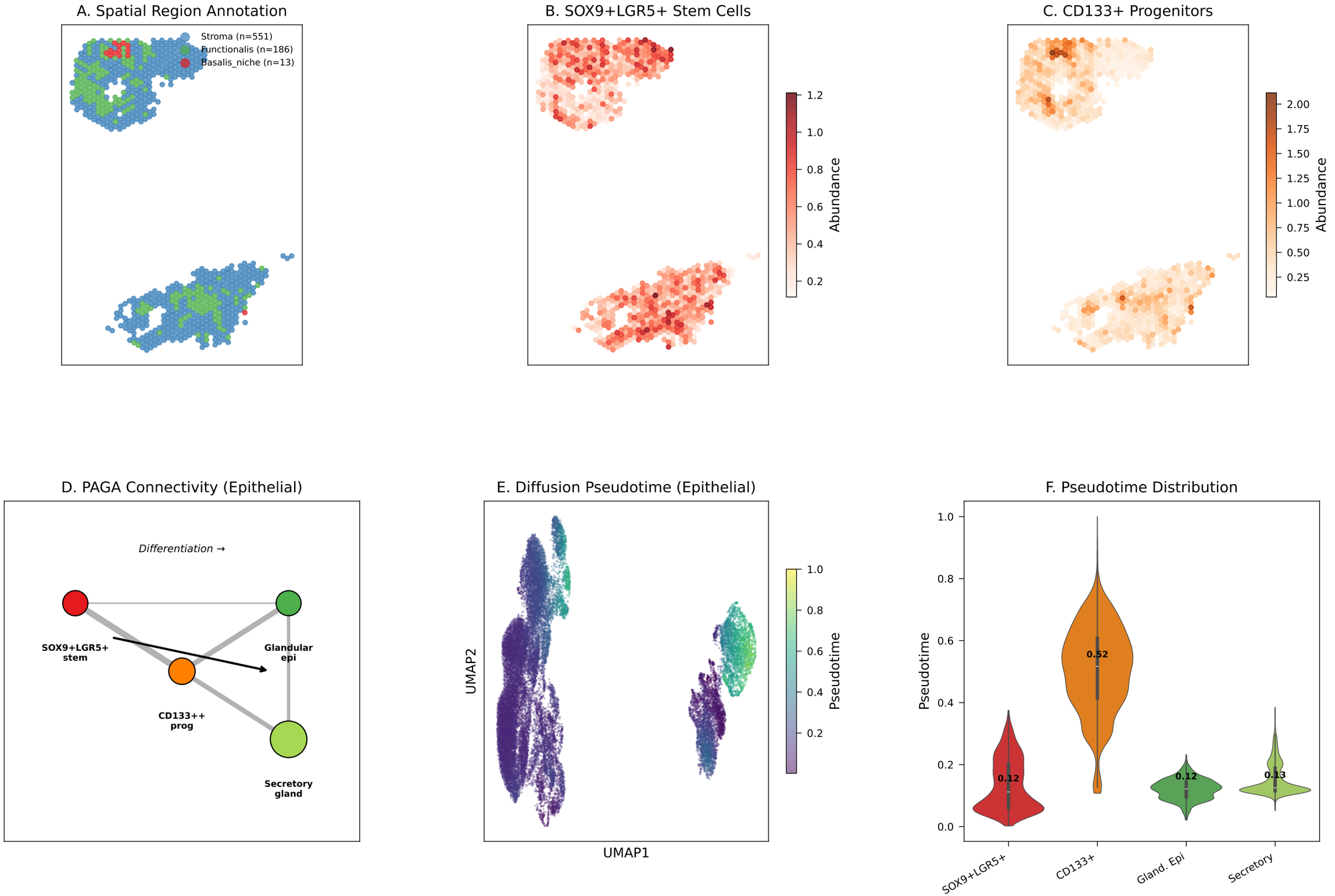
A. Spatial Region Annotation
Stroma (n=551)
Functionalis (n=186)
Basalis_niche (n=13)
B. SOX9+LGR5+ Stem Cells
Abundance
1.2
1.0
0.8
0.6
0.4
0.2
C. CD133+ Progenitors
Abundance
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
D. PAGA Connectivity (Epithelial)
Differentiation →
SOX9+LGR5+ stem
Glandular epi
CD133++ prog
Secretory gland
E. Diffusion Pseudotime (Epithelial)
UMAP2
UMAP1
Pseudotime
1.0
0.8
0.6
0.4
0.2
F. Pseudotime Distribution
Pseudotime
1.0
0.8
0.6
0.4
0.2
0.0
0.12
0.52
0.12
0.13
SOX9+LGR5+
CD133+
Gland. Epi
Secretory